Towards the design of a Real-Time processing System for Forcecardiography Signals

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***Abstract*—Forcecardiography (FCG) is a novel technique for the non-invasive acquisition of cardiac and respiratory induced chest wall vibrations that has been demonstrated using innovative, non-invasive force sensors. FCG sensors offer richer informational content by capturing infrasonic and audible vibrations that are not captured by other cardio-mechanical monitoring techniques like seismocardiography (SCG). However, the extraction of relevant vibrational components from FCG data requires dedicated processing that has been proposed and only demonstrated with off-line processing. This paper analyses the design trade-offs for the implementation of a compact processing system for the real-time extraction and wireless transmission of FCG components, which could enable home-care and telemedicine applications. The design target is an ARM/FPGA SoC board for acquiring, processing, and wireless transmission of the processed data. Preliminary results indicate that efficiently extracting various frequency bands in real time with a latency of less than 200 ms is feasible.**

Keywords—forcecardiography (FCG), seismocardiography (SCG), electrocardiography (ECG), hardware implementation, real-time, FPGA, System on Chip.

I. INTRODUCTION

Cardiac monitoring has always been a crucial part of medical diagnostics and care, aiming to understand and treat cardiovascular diseases. Historically, the mechanical vibrations generated by cardiac contractions have been assessed through palpation and auscultation, methods that while fundamental, are subjective and prone to restrictions. Advances in technology, and specifically in microelectromechanical systems (MEMS) technology has made researchers revisit non-invasive sensor-based approaches like seismocardiography (SCG) [1], [2], employing small and lightweight inertial sensors, placed on the chest of the subject. SCG is based on small accelerometers and is currently considered the cornerstone of cardiomechanical monitoring. SCG research has focused on analyzing the temporal relationship of its peaks and valleys, identifying fiducial markers linked to specific cardiac events [2]-[8]. In addition, numerous studies attempted to increase the accuracy of SCG by removing noise and motion artifacts, like normalized least mean square adaptive filters [9], empirical mode decomposition [10], [11], adaptive recursive least square filters [12], and using two tri-axial accelerometers in various configurations [13].

Forcecardiography (FCG) is a novel, non-invasive technique, used to measure cardiorespiratory vibrations [14], [15]. Unlike SCG, which primarily captures acceleration-based signals, FCG uses force sensors to measure the weak forces impressed on the chest wall by the mechanical activity of heart and lungs. This approach enables the simultaneous acquisition of a signal related to the respiratory activity, referred to as forcerespirogram (FRG), and a signal containing both infrasonic and audible cardiac-induced vibrations, which is the actual Forcecardiogram. This cardiac signal contains a high-frequency infrasonic component named HF-FCG (7 Hz – 30 Hz), which shares a high similarity with SCG, and an audible component named HS-FCG (30 Hz – 200 Hz) which corresponds to typical heart sounds recorded by an electronic stethoscope.

In particular, the first derivative of the HF-FCG component has shown the highest similarity to the accelerometric SCG signal and allowed accurate localization of the aortic opening fiducial markers that are usually identified in SCG signals [16]. All cardiac FCG components have shown a very high correlation to heartbeats, thus allowing accurate estimation of instantaneous heart rates. Moreover, the amplitude and morphology of these components have been found to be modulated by the respiratory activity [17].

FRG and cardiac FCG components can be extracted from the raw FCG sensor signal via specific processing procedures, which have been implemented offline. This paper presents the first real-time hardware implementation of a FCG signal processing system, transforming the offline approach into a real-time demonstrator useful for dataset collection.

The proposed system converts the analog signal from the sensor into a digital format and utilizes a System on Chip (SoC) development board to process and filter the signal, extracting its components. The Analog to Digital Converter (ADC) feeds the processing system with 1 kSamples per second, which corresponds to 24 kbps. The processing system exploits both the ARM core and the FPGA to process the data in real-time. Furthermore, the extracted FCG components are transmitted to an external microcontroller board via SPI (serial peripheral interface), which wirelessly sends the data to a server for visualization purposes or further analysis, if so desired. The chosen SoC development board is the Zybo Z7 from Digilent, which integrates a dual-core ARM Cortex-A9 processor with AMD 7-series Field Programmable Gate Array (FPGA) logic, while the chosen microcontroller board is the Raspberry Pi Pico W which features a dual-core Arm Cortex M0+ processor and offers wireless connectivity options.

The paper is organized as follows: the input data and the general workflow are presented in section II, performances of the various components are shown in section III, and conclusions are drawn in section IV.

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II. Input Data and Processing Workflow

A. Forcecardiography reference signals

The FCG signals used in this study had previously been acquired in [15]. To this aim, a lightweight FCG sensor (diameter of 30 mm) had been attached to the chest of healthy volunteers via medical adhesive tape, around the point of maximal impulse (fifth intercostal space on the midclavicular line), and secured via an elastic band wrapped around the thorax. FCG signals and a reference electrocardiographic (ECG) signal had been acquired simultaneously from the healthy volunteers, during quiet respiration, while comfortably sitting on a chair, leaning against the seatback and keeping their back straight. The signals had been digitized at 10 kHz with 24-bit precision via a NI-USB4431 data acquisition board. The ECG and FCG signals as acquired in [15], along with their corresponding frequency components, are shown in Fig. 1.

B. Processing chain

The workflow of the system is presented in Fig. 2. The analog version of the test signal discussed in II.A is converted to digital by an ADC. The ADC transmits this digital data, consisting of 12 to 32 bits per sample (depending on the ADC used), to the FPGA development board via SPI, with a data rate of 1kSps.

On the FPGA, a Savitzky–Golay filter, implemented using high-level synthesis, extracts the low-frequency force-respirogram (FRG) from the input signal, following the approach of [15], [16]. This respirogram is then subtracted from the input signal to obtain the force-cardiogram (FCG). The FCG is further processed using second-order bandpass IIR filters, implemented in software, to extract the HF-FCG and HS-FCG components. Finally, once the data is filtered, it is organized in packets with a Packet ID and CRC code that help maintain data integrity.

At this point the ARM processor on the Zybo Z7 issues an interrupt that triggers an external microcontroller, the Raspberry Pi Pico W, that is equipped with wireless connectivity. The microcontroller is configured as an SPI master, and once triggered, it initiates a transmission to receive the input data, the filtered output, the packet ID, and the CRC verification code. The Raspberry Pi Pico W is a dual-core microcontroller, with the interrupt service routine (ISR) tied to the interrupt from the Zybo, running on its first core. The received data is stored in a circular buffer which is shared between the two cores. The second core reads data from the buffer, combines it into larger packets, and sends it via UDP to a remote server connected to the same access point as the Raspberry Pi Pico W.

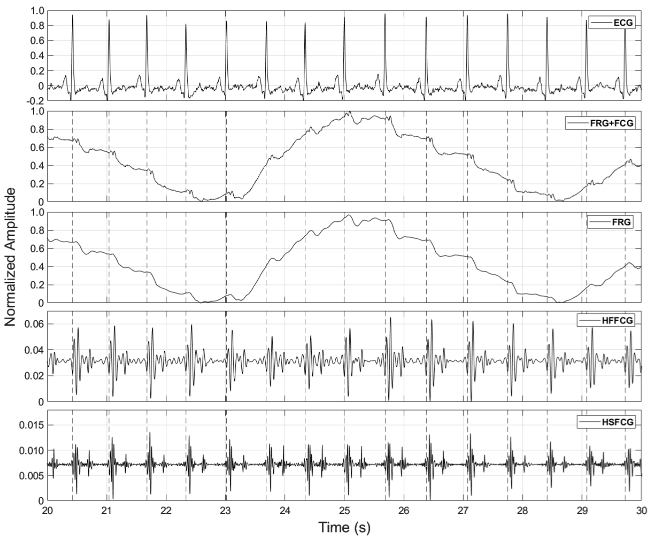

Fig. 1: ECG for reference, FRG+FCG (input signal), FRG (respiration), HF-FCG (7 Hz – 30 Hz), HS-FCG (30 Hz – 200 Hz).

III. Performances

A. ADC Selection

The raw analog signal from the FCG sensor must be converted into a digital format before it can be processed and filtered. The quality of the processed data largely depends on this initial step (ADC) that however poses important trade-offs between the desired resolution and accuracy of the detected signal and the power dissipation and speed of the hardware processing system. To this purpose, the following ADC solutions and configurations have been evaluated: *xADC* 12bit, *STM Nucleo-H755ZIQ* 16bit, *PmodAD5* 24bit, and *TI ADS1282EVM 32*bit.

xADC Integrated into the Zybo Z7 processing board, offers a dual 12-bit ADC with a maximum sampling frequency of 1 MSPS, and an input voltage range from 0V to 1V. Since the xADC is built into the CPU, it can be accessed directly and requires no protocol to transfer the digital data to the Zybo processing cores. Its power dissipation is around 10 to 50 mW depending on the operation mode and the clock frequency.

STM Nucleo-H755ZIQ The 16-bit ADC is available with the dual-core microcontroller development board from STMicroelectronics that features an Arm Cortex-M7 core @480 MHz and a Cortex-M4 core running at up to 240 MHz. It dissipates around 1.4 W at runtime. It supports multiple ADC channels, with resolutions of up to 16 bits achievable through hardware oversampling and averaging, and a maximum sampling rate of 3.6 MSPS. Being an external board, after the data stream is converted to digital, it is organized into packets, assigned a packet ID, and a CRC verification code is computed. The packets are then transmitted via SPI to the Zybo board. If the expected Packet ID is not received, or in case of a CRC mismatch, the Zybo asks for the packet again.

PmodAD5 It is an external ADC module compatible with the Zybo Z7, and built around the Analog Devices AD7193, a low-noise, 4-channel, 24-bit sigma-delta ADC. It has a resolution of 24 bits, a 4.8 KSPS sampling frequency, and consumes around 14 mW at runtime. It has an input voltage range from -2.5V to 2.5V and offers four differential or eight single-ended channels. It can be directly connected to the Zybo Z7 via the Pmod port and uses SPI communication for seamless integration with the processing board (configuration and data transfer).

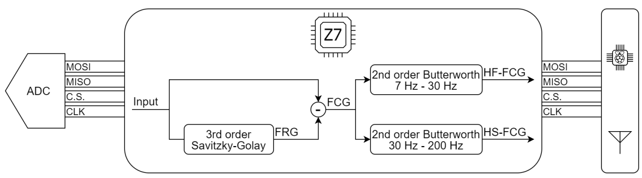

Fig. 2: Overview of the proposed system and processing chain.

TI ADS1282EVM It is a high-performance evaluation module designed to house the ADS1282, a precision 32bit delta-sigma ADC. The sampling frequency is configurable, with a maximum rate of 4 KSPS. The module supports differential input voltages within the range of ±2.5V. Additionally, it features ultra-low noise performance and programmable gain amplifiers. This high-performance ADC dissipates 650 mW, requiring symmetrical external power supply. Like the other ADCs, the ADS1282EVM can communicate with the Zybo Z7 via an SPI interface.

To evaluate the performance of the available ADCs, the FCG signal (shown in Fig. 1) was uploaded to an Arbitrary Waform Generator (AWG). Each ADC was then used to sample the signal at 1 kHz, the digital outputs were recorded, and the Mean Squared Errors (MSE) with respect to the original input were computed. The xADC, with only 12-bit resolution, performed reasonably well but exhibited significant random spikes scoring an MSE equal to 3.9×10^{-5}. The 16-bit ADC on the STM Nucleo-H755ZIQ delivered very good performance, closely matching the input signal and reaching an MSE of 2.1×10^{-5}. The PmodAD5 with a 24-bit resolution, and the TI ADC with a 32-bit resolution demonstrated exceptional performance, producing signals nearly identical to the original, and scoring 5.1×10^{-6} and 5.0×10^{-5}, respectively. The results allign with expectations except for the ADC by TI. While the TI ADC offers a very accurate signal, it suffers from signal drift due to instabilities in the external power supply and high power dissipation. The PmodAD5 outperformed the STM ADC by a factor of four while also being a much more compact solution, thus making it the preferred ADC for this study.

B. *Filtering*

As shown in Fig. 2, the first applied filter is a third order Savitzky–Golay filter. This FIR filter has 151 symmetric coefficients and applies a polynomial smoothing approach centered on each data point, which effectively preserves the features of the signal thus extracting the FRG component. Since each data point requires 75 preceding and 75 following samples to compute the Savitzky-Golay filter output, the first computable output corresponds to the 76th sample. This output becomes available with the 151st input, as only then are all required samples available. Thus, the expected latency introduced due to the Savitzky-Golay filter is 75 ms. This filter has been described in C language and, via a High-Level Synthesis tool, converted into RTL code that can be synthesized and implemented on the FPGA, running at 60 MHz without pipelining and takes 2.5µs to complete.

At this point, the forcecardiogram (FCG) can be obtained as a simple subtraction of the FRG from the input data.

Zero-phase forward and reverse IIR filtering using the MATLAB function "filtfilt" has been shown to effectively extract the desired frequency components from FCG [15]. However, when the filtering takes place in real time without the knowledge of future inputs and outputs, backward filtering is impossible. To emulate this function and achieve near zero-phase distortion, each filter is implemented twice and performs only forward filtering. The two filters "HF-FCG", and "HS-FCG" are 2nd order Butterworth bandpass filters with passbands from 7 Hz to 30 Hz, and from 30 Hz to 200 Hz, respectively. The corresponding coefficients have been computed with the help of the MATLAB function "butter"

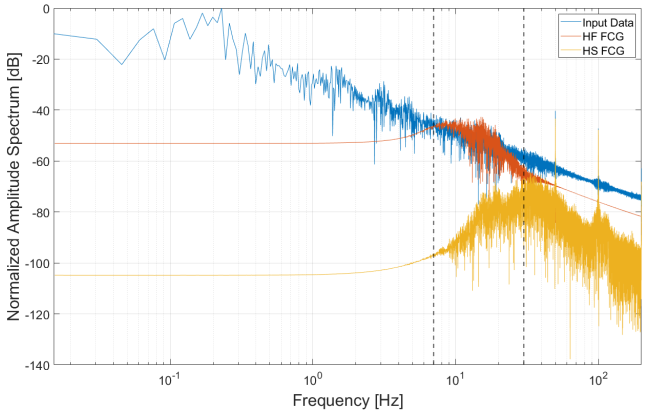

Fig. 3: Amplitude spectrum of the input/output signals. All the amplitudes are normalized to the peak frequency component of the input data.

and represented with 64-bit double-precision. The IIR filtering takes place on the ARM core which runs at 667 MHz.

The FFT analysis of the input (as sampled by the PmodAD5 ADC), the HF-FCG, and the HS-FCG signals are shown in Fig. 3. The frequency range between the two vertical dashed lines represents the passband that corresponds to the HF-FCG. Beyond the rightmost vertical dashed line, HS-FCG exhibits the highest amplitude, surpassing all other components. The spikes in the FFT plot at 50Hz and its harmonics are caused by powerline interference and appear in the input and HS-FCG signals, as their frequency ranges include these components.

The performance results of the system in the time domain for the same signal and timeframe as in Fig. 1, are shown in Fig. 4. The ECG and the vertical dashed lines corresponding to its peaks are included again as a reference. The same FCG input signal from Fig. 1 is sampled using the PmodAD5 ADC. The respiration signal, along with the HF-FCG, and HS-FCG components, are then extracted using the proposed filters. As shown in Fig. 4, the input, FRG, and HF-FCG signals closely follow the original data published in [15]. While the HS-FCG signal also follows the heartbeat, its noise floor appears significantly higher than in Fig. 1. This is because unlike the corresponding waveform in Fig. 1, the extracted HS-FCG signal from the proposed system includes 50 Hz interference, as also confirmed by the FFT analysis in Fig. 3.

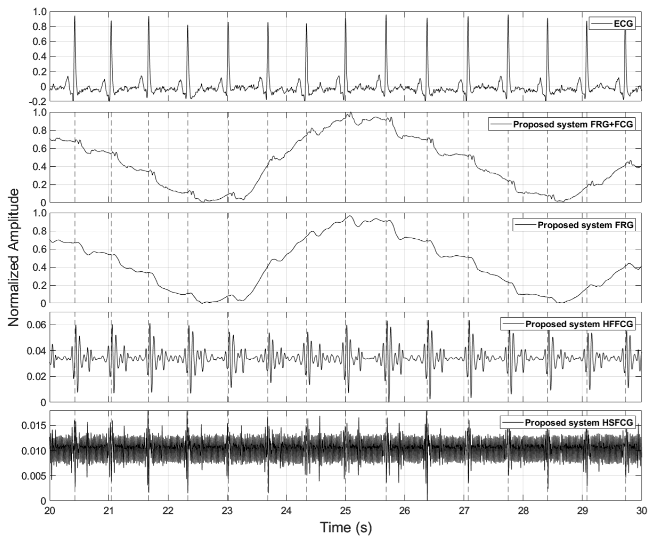

Fig. 4: ECG for reference, FCG sampled by the PmodAD5 ADC, FRG, HF-FCG, and HS-FCG computed by the proposed system.

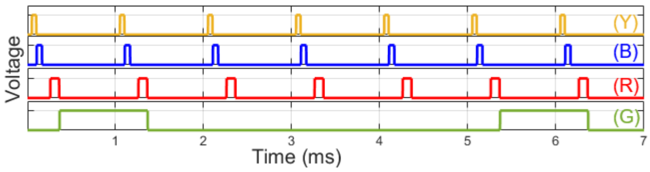

Fig. 5: Timing of the processing phases. (Y): ADC conversion, (B): filtering, (R): SPI transmission Zybo-R.Pi W, (G): WiFi transmission R.Pi W - Remote server.

C. Timing

An internal timer is configured to issue an interrupt every millisecond. With each interrupt, the ADC is activated to read its input and transmit the data via SPI to the Zybo Z7 board, a process that takes approximately 50 µs, and corresponds to the yellow trace (Y) in Fig. 5. Once the ADC transfer is complete, the filters update their input values and process the sampled signal, a task that takes around 60 µs, shown by the blue trace (B) in the figure. Following the filtering step, the processed data is sent via SPI to the microcontroller. With an 8 MHz SPI clock and 256 bits of data to transmit, the clean transmission time is 32 µs. However, accounting for the time-lag between each byte transmission and the time required for both boards to confirm the end of transmission, the total time amounts to approximately 100 µs, represented by the red trace (R). The time gap between the blue and red traces corresponds to the processing needed to assign a packet ID and calculate the CRC, taking an additional 100 µs.

The green trace (G) illustrates the time required for WiFi transmission. While there is some overlap with other signals, this does not cause conflicts as the WiFi transmission operates on a dedicated core of the microcontroller. As shown in Fig. 5, the microcontroller batches five consecutive packets and transmits them together, sending each batch three times to mitigate potential packet loss. Under normal network conditions, this process takes approximately 1 ms.

D. Power and FPGA resources

The estimated power dissipation of the synthesized netlist for the processing FPGA board is 1.588W. The design utilizes only two DSP blocks, specifically for implementing the FIR Savitzky-Golay filter. Additionally, 8,144 slice registers and 7,314 slice LUTs are employed in the design, most of which are used to connect the processing system (ARM Cortex) to the programmable logic (FPGA fabric). Overall, the resource utilization is minimal compared to the total available resources on the Zybo Z7 FPGA board, allowing for future additions and improvements.

IV. Conclusions

This paper presents the first system capable of deriving real-time information from a forcecardiography sensor and wirelessly transmitting it to a remote server. The system effectively isolates the respiratory component and extracts cardiac vibrational components of the FCG signal. This paves the way for the development of a personal device for continuous long-term monitoring of cardiorespiratory health, which could enable home-care telemedicine. Future work will focus on enhancing the filtering process and identifying fiducial markers from the extracted signals in real time.

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